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Comparative Analysis of Machine Learning Models for Crop Recommendation with a Deployed Random Forest Based Decision System

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To enhance agricultural output and sustainability, it is necessary to choose the most appropriate crop depending on the characteristics of soil and environmental situation. The conventional methods of crop selection are primarily based on the experience and traditions of farmers that are not always applicable to the alteration of soil nutrients and the climate conditions. In a bid to overcome this shortcoming, this research paper will suggest a machine learning-based crop recommendation model with a publicly accessible Kaggle dataset consisting of soil nutrients (Nitrogen, Phosphorus, and Potassium) and the environment (temperature, humidity, rainfall and pH). The evaluation of six algorithms of machine learning, such as Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (RBF), Random Forest, Gradient Boosting and XGBoost, had been conducted using proper data preprocessing and feature engineering tools. The random forest classifier performed best among them with the accuracy of 99.32, precision of 99.35 and the recall of 99.32. In order to guarantee model reliability, 10-fold stratified cross-validation was conducted, the average result was 99.59% with a standard deviation of 0.38 as a result. The most effective Random Forest model was combined with an interactive interface in Streamlit and FastAPI allowed the scalable inference on the backend. The entire system was implemented on Hugging Face platform where the users could receive real-time crop recommendations via a web-based application. The gap existing between research and real agricultural practices has been filled in this work as it offers a practical tool that is deployable and available to assist in a sustainable farming system.

1.1 INTRODUCTION

The agriculture sector is a critical element in the maintenance of food security in the world and in the economic growth of most nations especially the developing ones [1]. The choice of the right crop that suits a given soil type and environmental status is imperative in ensuring that the crop yields optimally and wastage of resources is reduced to a minimum [2]. Historically, the process of crop selection has been dependent on the experience of the expert and past farming methods [3]. Nevertheless, the traditional methods tend to overlook the dynamism of the soil nutrients, the factors in the environment as well as the change in the climatic conditions that greatly affect the productivity of agriculture.

Due to the development of new technology, information-based methods have been gradually gaining importance in the agricultural sector. Machine learning methods offer the effective means of analysis of extensive data on agriculture and revealing the patterns that would not be instantly noticeable with the help of conventional algorithmic methods [4]. Machine learning models have the potential of predicting the best crops to be used in certain conditions by examining the soil properties, like Nitrogen, Phosphorus, and Potassium levels, as well as environmental variables, including temperature, humidity, rainfall, and pH [5]. This will allow farmers and other stakeholders in the agricultural sector to make better decisions regarding the crop determination and control.

A number of research works have addressed machine learning algorithmic implementation in crop recommendation system. Efforts like Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Gradient Boosting, and XGBoost have been extensively used in predicting the agricultural activities [6][7]. Some of such techniques include tree-based ensemble models like Random Forest that tend to perform well because they can work well with complex data and minimise overfitting without compromising prediction accuracy [8].

It is also important that feature engineering helps to enhance the performance of machine learning models [9]. The model performance can be greatly improved by techniques that include the generation of statistical features, making ratios of features, transformations as well as the introduction of polynomial interactions. In addition, the incorporation of machine learning models into user-friendly solutions will facilitate real-time recommendations, which will bridge the scholarly study and practical use in agriculture fields [10].

In spite of the increased attention to machine learning-based crop recommendation systems, there are still a number of limitations. The majority of available studies are focused on enhancing the accuracy without introducing the use of the robust validation techniques like the k-fold cross-validation. Also, comparisons between various algorithms are frequently displayed, but the most efficient models are not commonly implemented into any workable system that can be easily accessed by farmers. It also has no combined frameworks that include effective data preprocessing, feature engineering, and deployment architecture. Consequently, it is still evident that there is a high demand of an evaluable and implementable crop recommendation system that has the potential to deliver scalable and real time decision support to farmers. Our contributions are,

1. Comparison among six models on crop recommendation based on machine learning concepts and rigorous data pre-processing.
2. The observation that Random Forest provides the highest predictive accuracy and cluster separation compared to the others.
3. Applied 10-fold CV; Random Forest got 0.9959 +/- 0.0038 accuracy; which is strong in generalization.
4. Calculated the model and embedded it within a Streamlit web app as real-time recommendations.
5. Implemented the system with Fast API and Hugging Forest, which allows agricultural decision support to be scaled and made accessible.

The remainder of this paper is organized as follows. **Section 1.2** discusses related works in the area of Crop Recommendation. The machine learning-based models and datasets are given in **Section 1.3**, followed by discussions on the classification performances of the models in **Section 1.4**. Finally, **Section 1.5** concludes the paper.

1.2 RELATED WORK

More recently, machine learning has entered agriculture to help determine the type of crops to be planted and how much yield should be expected. Various ML methods have been applied to optimize crop selection based on soil health and environmental conditions. Prof. S et al.[11] used regression and classification methods on how soil nutrients and climate conditions feed into predictions that determine which crop would perform well. Similarly, S. Mondal et al. [12] demonstrated that crop recommendations can become more accurate with ensemble learning method. People have also made head-to-head comparisons between different ML models in an attempt to find the best models. S. Sharma et al. [13] investigated the performance of Logistic Regression, K-Nearest Neighbors, and Support Vector Machines against crop yield prediction, mentioning that tree-based ensembles are able to adapt high-dimensional data more effectively. Ainaa Hanis Zuhairi et al. [14] have identified Gradient Boosting and Random Forests as outperforming linear models for crop recommendation tasks. Feature engineering has been found to be a driver of improved performance. G. Yokoyama et al. [15] emphasized that construction of interaction features, ratio-style soil nutrient metrics, and logarithmic transformations can increase predictive accuracy. S. P. Harshitha L et al. [16] employed similar techniques and realized significant improvement in both precision and recall. Finally, hybrid feature approaches which combined statistical aggregation and polynomial interactions yielded more robust models. None H et al. [17]. The existing literature reveals the application of ML models being implemented in the form of interactive tools for the farmers [18] demonstrated crop recommending systems implemented with the use of ML, which are performed through web-based applications, providing real-time assistance to the farmers. Yaganteeswarudu Akkem et al. [19] implemented a crop recommending system based on the Random Forest approach implemented in Streamlit, which performed well in terms of accuracy along with a pleasant interface for the users. Among various ML models compared, the efficacy of the Random Forest approach is evident due to the ability to overcome overfitting along with accommodating a large set of features [20]. Overall, it is evident from all said efforts that a combination of effective feature engineering and ensemble learning with Random Forest, and even implementing them interactively, can form a reliable and viable process with regard to modern crop recommendation systems. All these efforts, taken together, form a strong foundation for this particular study, where a comprehensive assessment is carried out with six different models in machine learning, accompanied by real-time recommendations employing the best random forest model.

1.3 METHODOLOGY

This section provides the proposed model of crop recommendation by a machine learning-based application with machine deployment. The general flow of work is presented in Figure 1 that describes the entire process starting with data preprocessing and ending with the process of prediction in real-time and deployment of the system. The steps include preprocessing soil and environmental information, and training several machine learning models to be used in crop prediction. Performing the best models are then picked and incorporated into a final recommendation system to enhance accuracy and reliability. Lastly, a web-based application is used to deploy the trained model to give farmers real time crop recommendations depending on the input conditions.

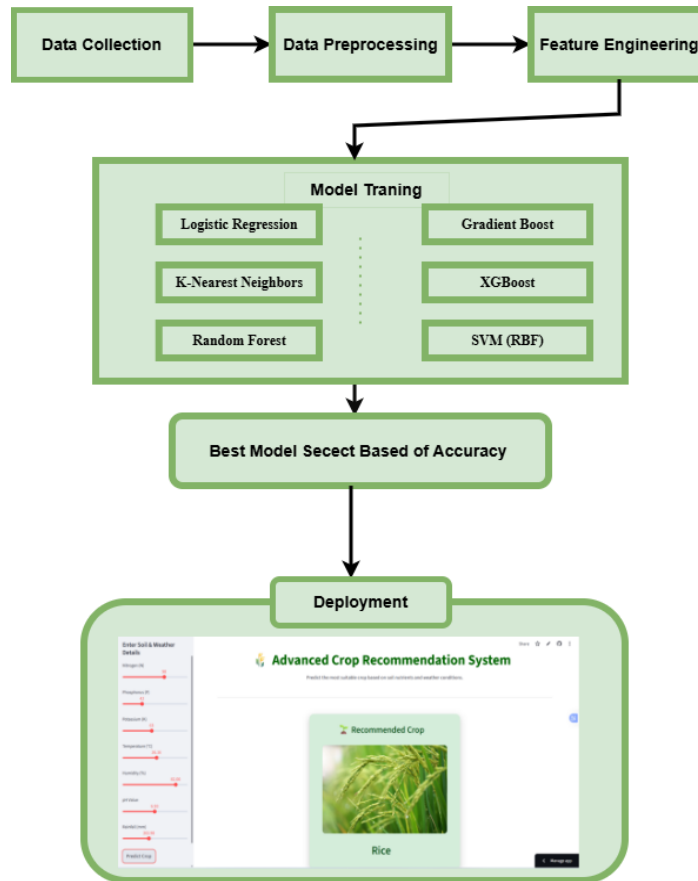


Figure 1.1: Research Workflow Diagram

1.3.1 Dataset Description

The dataset on crop recommendations was taken from kaggle [21], contains soil and climatic parameters to identify the most appropriate crop to be grown. It contains seven numerical variables, namely Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity levels, pH levels, and rainfall, representing soil quality and climatic parameters. The dependent variable is the crop class, which includes varied categories of crops. The data set is properly organized with no missing values and can be effectively used for

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multi-class classification problems to identify quality crop selection using machine learning algorithms based on soil and climatic parameters.

1	Nitrogen	Phosphorus	Potassium	Temperatur	Humidity	pH_Value	Rainfall	Crop
2	90	42	43	20.879744	82.002744	6.5029853	202.93554	Rice
3	85	58	41	21.770462	80.319644	7.0380964	226.65554	Rice
4	60	55	44	23.004459	82.320763	7.8402071	263.96425	Rice
5	74	35	40	26.491096	80.158363	6.9804009	242.86403	Rice
6	78	42	42	20.130175	81.604873	7.6284729	262.71734	Rice
7	69	37	42	23.058049	83.370118	7.0734535	251.055	Rice
8	69	55	38	22.708838	82.639414	5.7008057	271.32486	Rice
9	94	53	40	20.277744	82.894086	5.7186272	241.97419	Rice
10	89	54	38	24.515881	83.535216	6.6853464	230.44624	Rice

Figure 1.2: Datasets Visulization

1.3.2 Data Preprocessing

Data Preprocessing is achieved through various processes to ensure the quality of the data before modeling. First, the data is analyzed for the existence of any missing values. Missing numerical values in the data are replaced using the median imputation method. The reason why the median imputation process is used is that it can help in the reduction of the effects posed by extreme values. Secondly, dealing with numerical outlier data (N, P, K, temperature, humidity, pH, rainfall) is done using the Interquartile Range (IQR) technique. This technique removes data points beyond the desirable range of IQR. It ensures there is no effect caused by this data during the learning phase of modeling. Thirdly, the category labels are changed into numeric representations using a procedure called “label encoding.” These new class labels are stored under “new class labels” and will be used later for other tasks like the confusion matrix and t-SNE. After the data is preprocessed and the creation of the feature variables, the data is split into a train data and a test data with an 80:20 split using stratified sampling. The stratified sampling ensures that the data in the train data and the test data is equally distributed.

Table 1.1: Data Preprocessing Steps

Step	Technique Used	Purpose
Missing Value Handling	Median Imputation	To handle missing numerical values while minimizing the effect of extreme values

Outlier Treatment	IQR-based Clipping	To reduce the influence of extreme values in soil and environmental features
Train–Test Split	80:20 Stratified Split	To preserve class distribution and ensure unbiased model evaluation

1.3.3 Feature Engineering

In order to improve the predictive ability of models, advanced feature-engineering techniques are applied. The Ratio Features include the calculation of the values of the following ratios: Nitrogen/Phosphorus (N/P), Phosphorus/Potassium (P/K), and Nitrogen/Potassium. Statistical details about the mean and standard deviation values for N, P, and K can be achieved in order to provide information about soil fertility [24]. Functions related to rainfall and humidity variables are transformed in order to make the variance stable and reduce skewness [23]. On the other hand, polynomial interaction terms with a degree of two are formed in order to define the relationship between the variables pertaining to the soil and the environment [26].

Table 1.2: Feature Engineering Techniques

Feature Type	Technique Used	Purpose
Ratio-Based Features	N/P, P/K, N/K Ratios	To capture soil nutrient balance
Statistical Features	Mean & Standard Deviation of N, P, K	To provide aggregated soil fertility information
Logarithmic Transformation	$\log(1 + \text{rainfall})$, $\log(1 + \text{humidity})$	To reduce skewness and stabilize variance
Polynomial Interaction	Degree-2 Interaction Features	To model non-linear relationships among features

1.3.4 Models Selection

To compare machine learning methods in crop recommendation, six classifiers were selected. The reasons for choosing these classifiers include the characteristics of the respective algorithms and the algorithms' applicability to the problem of multi-class classification on well-structured agricultural datasets.

- **Logistic Regression:** A reason we could include Logistic Regression as a simple linear classifier is the fact that it is a fast and easy-to-understand approach. This approach allows the estimation of class probabilities using a linear boundary. This helps set a standard for comparison between more complex techniques.
- **K-Nearest Neighbors:** K-Nearest Neighbors is a distance-based algorithm that makes a prediction using the majority of its K nearest samples from the data space. This algorithm was introduced to include local properties of the data and to evaluate the appropriateness of instance-based learning on soil and environmental features.
- **Random Forest:** A random forest is essentially a combination of decision trees. It encompasses the generation of lots of decision trees and combining their predictions. This helps control overfitting. Additionally, it works well with several features. Moreover, random forest is capable of dealing effectively with non-linear relationships. Another important thing is that it is able to offer the measure of the features' importance. This is advantageous for agricultural datasets.
- **Gradient Boost:** Gradient Boosting learns models one after another, with each successive model trying to correct the errors of the preceding models. This was selected as a way of evaluating the ability of boost methods in improving their predictions for crop recommendation problems.
Gradient Boosting learns models one after one, with each successive model trying to correct the errors of the preceding models.
- **XGBoost:** This algorithm's full form stands for Extreme Gradient Boosting, and it goes by the name of 'XGBoost'. It acts as a boosting algorithm. It has been observed to be very efficient with systematic regularization to reduce overfitting, which makes it highly suitable for use on large data sets, such as this dataset found to be relevant to agriculture.
- **SVM (RBF):** One of the striking models used in supervised learning to work effectively with non-linear data is the Support Vector Machine (SVM) that uses the RBF (Radial Basis Function) kernel. The RBF kernel uses input features to project them into a higher dimensional space, which allows the model to form complex decision boundaries. It is extensively applicable to both classification and regression problems and it is efficient on small to medium-sized data.

1.3.5 Model Evaluation

We applied six machine learning algorithms on the testing dataset. To judge the efficiency of these machine learning algorithms in classifying, accuracy, precision, recall, F1 measure, and weighted F1 measure were used. To compare the efficiency of different classes in separating each other and performance of their features, confusion matrices were used, and t-SNE plots were derived. In comparison with other machine learning algorithms, accuracy was high in Random Forest.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$Weighted\ Precision = \sum_{i=1}^C \frac{n_i}{N} \cdot \frac{TP_i}{TP_i + FP_i} \quad (2)$$

$$Weighted\ Recall = \sum_{i=1}^C \frac{n_i}{N} \cdot \frac{TP_i}{TP_i + FN_i} \quad (3)$$

$$Weighted\ F1 - Score = \sum_{i=1}^C \frac{n_i}{N} \cdot \frac{2TP_i}{2TP_i + FP_i + FN_i} \quad (4)$$

In the multi-class setting, we applied weighted means to handle imbalanced data. The confusion matrices were helps to understanding overall performance measures, whereas t-SNE graphs gave us insights into how the features are distinguished. As expected, Random Forest gave us the best performance with respect to test accuracy.

1.3.6 System Deployment

observed that the Random Forest classifier is the most accruing model and thus has been used for building a practical model for recommending a crop. For practical usability of the proposed model, a user-friendly GUI has been designed using Streamlit. This GUI allows users to enter values of soil nutrients (N, P, K) and environmental factors like temperature, humidity, rainfall, and pH. The Streamlit app sends user inputs to the trained Random Forest model using a FastAPI-based backend, which is efficient and scalable. The model predicts the result in real-time, which includes visuals like the probability distribution of the classes and graphs. To make the model produce similar results, the same processing as performed on the model during training has been followed during deployment.

The entire application is hosted on the Hugging Face platform, which makes it easily accessible and scalable. This interactive hosting enables the model to be utilized as a decision-support system and will help these individuals make informed and data-driven decisions regarding the choice of the crops that will be cultivated.

1.4 RESULTS AND DISCUSSION

Systematic evaluation and comparison of the performance of the six machine learning models in crop recommendation was performed. Table 3 summarizes the results and shows the train and test accuracy, precision, recall, and F1-score of each of the models. The classifier with the best test accuracy of 99.32 showed the best precision, recall, and F1-score; therefore, suggesting that the Random Forest classifier is a strong predictor.

Table 1.3: Performance Analysis of Baseline Models Before Ensemble Learning

Model	Train Acc (%)	Test Acc (%)	Precision (%)	Recall (%)	F1-score (%)
Logistic Regression	98.92	98.64	98.69	98.64	98.63
K-Nearest Neighbors (KNN)	98.13	97.05	97.31	97.05	97.04
SVM (RBF)	98.58	98.64	98.70	98.64	98.63
Random Forest	99.00	99.32	99.35	99.32	99.32
Gradient Boosting	99.47	97.95	98.06	97.95	97.96
XGBoost	99.07	98.64	98.67	98.64	98.64

Each model has its confusion matrices presented in Figures 3 through 8, which show the performance of each model in the form of classes. The model of the Random Forest managed to record minimal misclassifications in all the classes of crops whereas other models like the Logistic Regression, SVM and the XGBoost showed some confusion among the similar crops. K-Nearest Neighbors and Gradient Boosting had a somewhat lower class separability, which indicates that they had relatively low overall

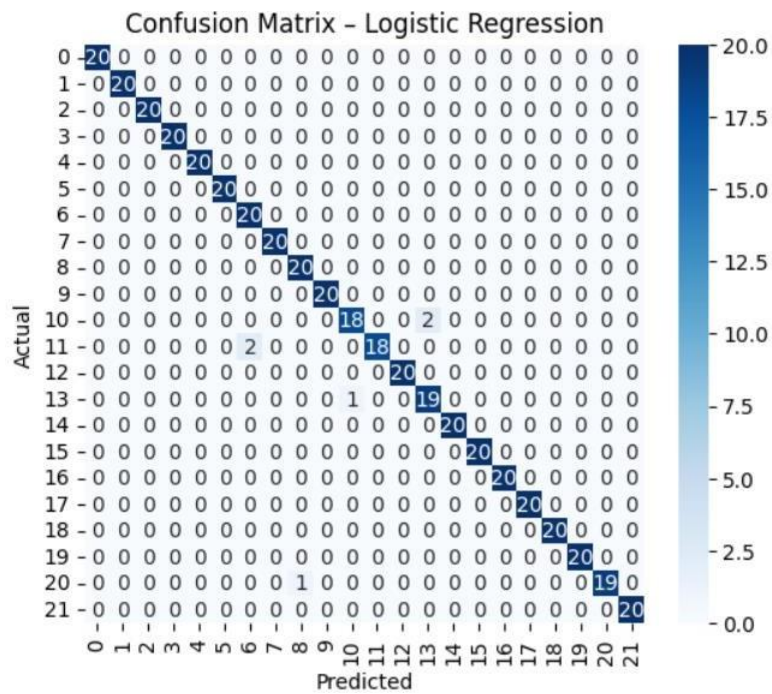


Figure 1.3: Confusion Matrix of Logistic Regression

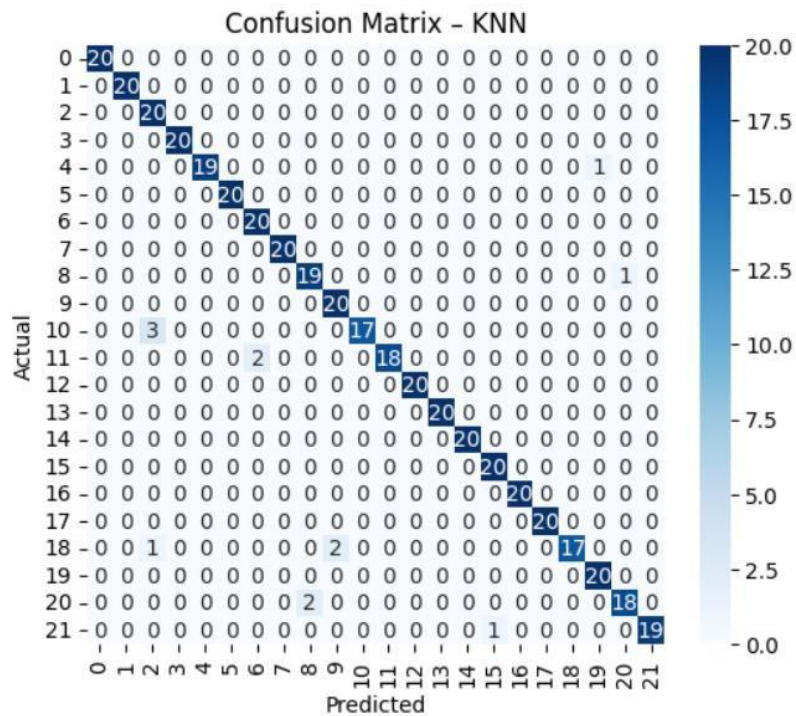


Figure 1.4: Confusion Matrix of KNN

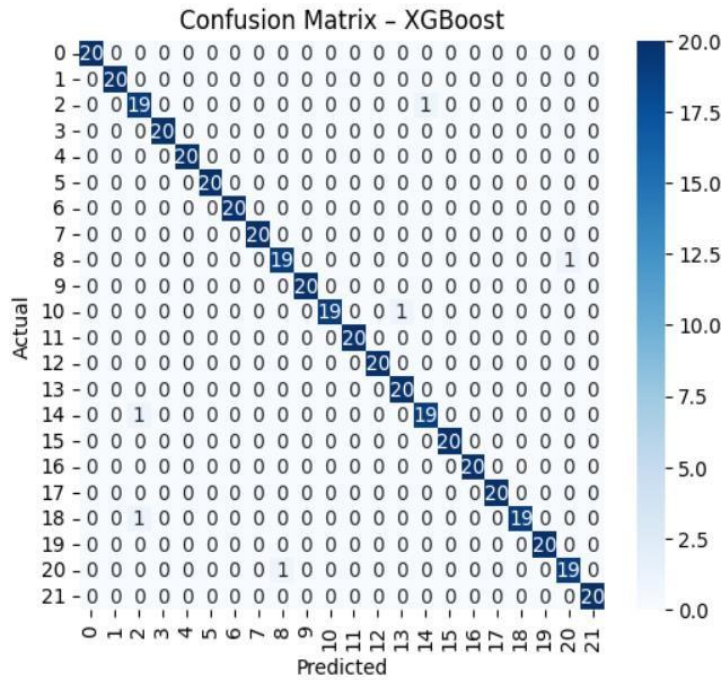


Figure 1.5: Confusion Matrix of XGBoost

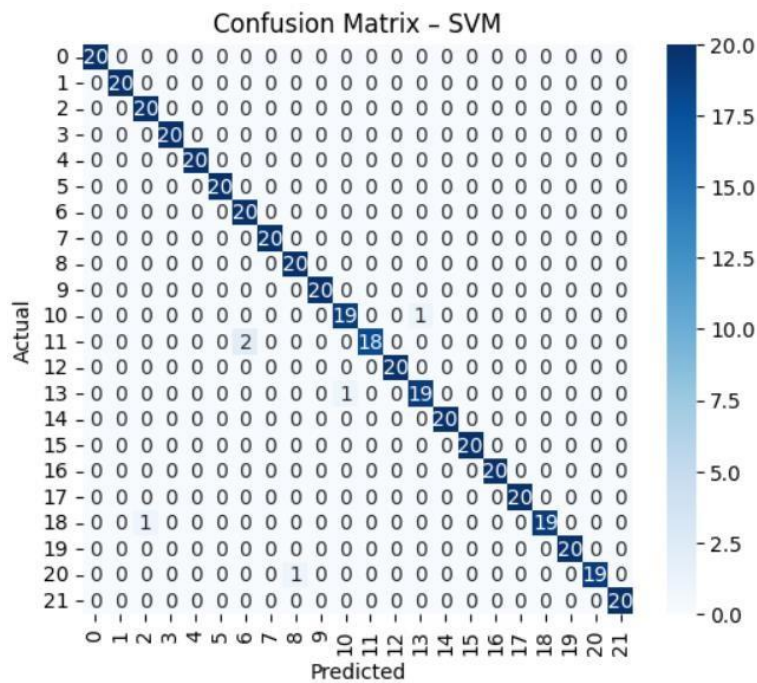


Figure 1.6: Confusion Matrix of Logistic Regression

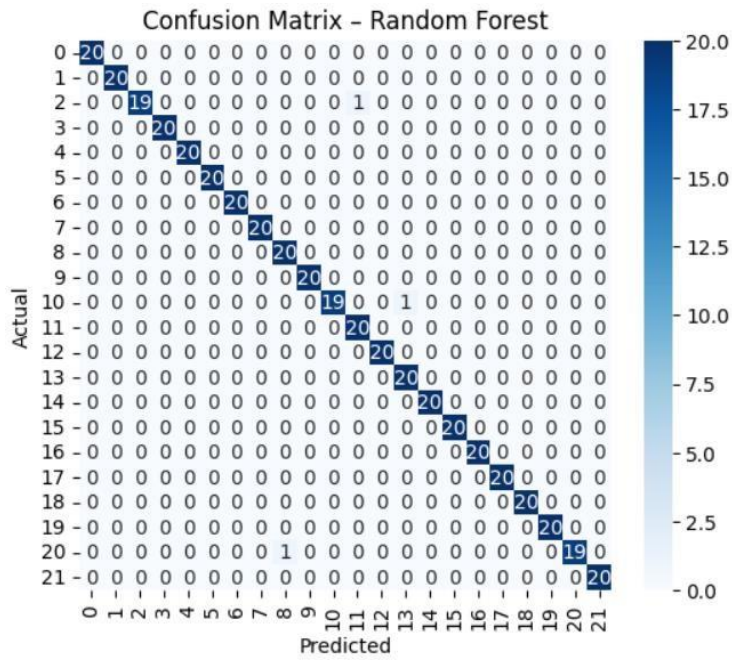


Figure1.7: Confusion Matrix of Random Forest

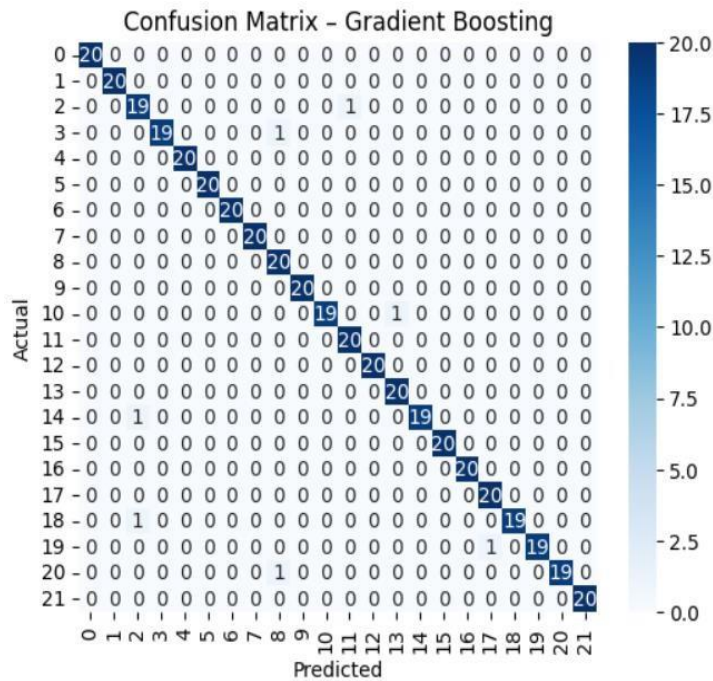


Figure 1.8: Confusion Matrix of Gradient Boosting

t-SNE representations of test data identified in Figures 9-14 show the quality of class separation of each model. Random Forest model yielded clear, well-articulated, and distinct crop groups, and had excellent visual separation when compared to other models. This unique clustering means that it can be used to describe complicated non-linear relationships of soil nutrients and environmental characteristics and therefore produce little overlapping classes and increased predictability. By contrast, there were observable overlapping clusters with Logistic Regression, KNN, SVM, Gradient Boosting, and XGBoost, especially between classes of similar soil and environmental properties, which are more consistent with their lower test accuracy and F1-scores in Table 3. Further evidence of the superiority of the Random Forest is also the t-SNE clusters that are clearer enough, which gives it the advantage of being chosen as the best-performing model to be deployed.

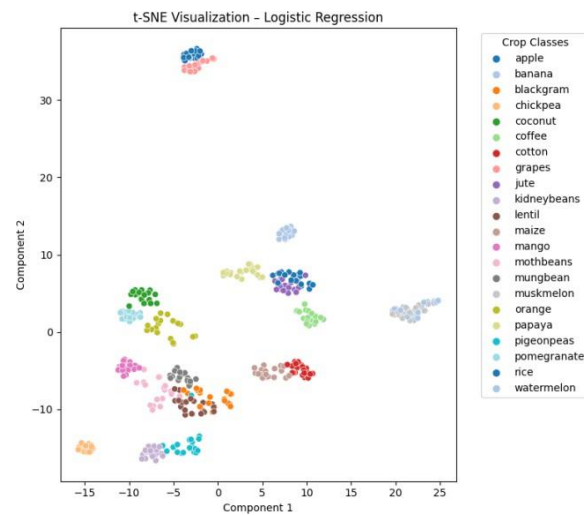


Figure 1.9: t-SNE of Logistic Regression

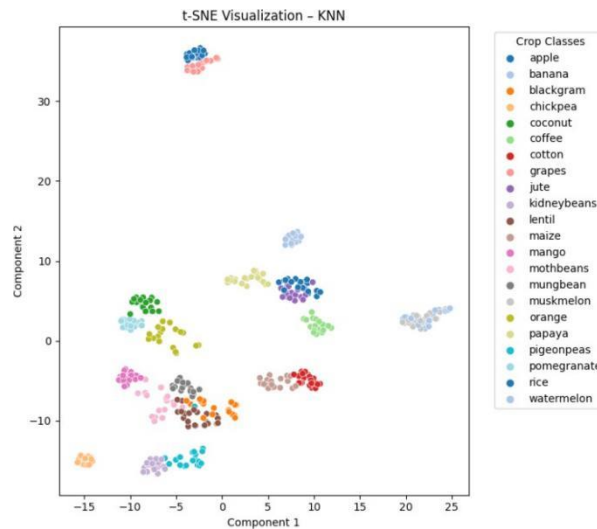


Figure 1.10. t-SNE of KNN

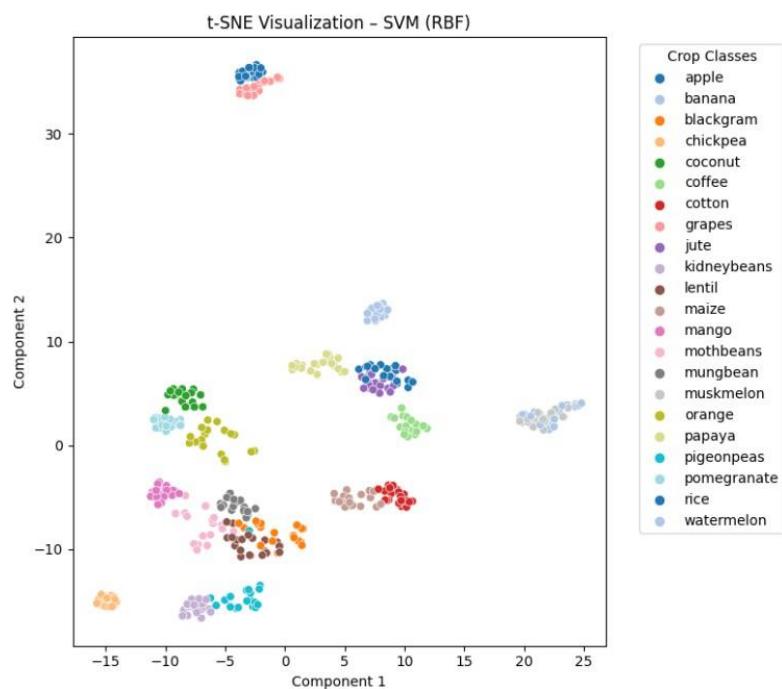


Figure 1.11: Confusion Matrix of SVM

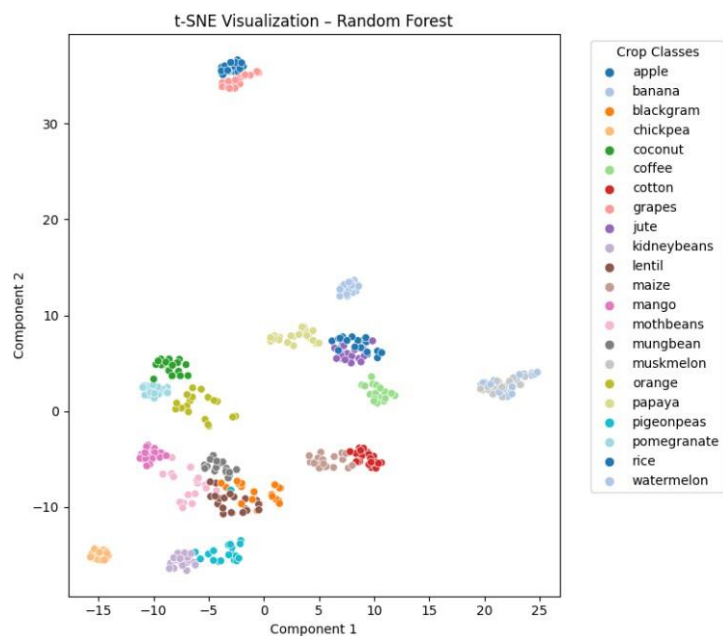


Figure 1. 12: t-SNE of Random Forest

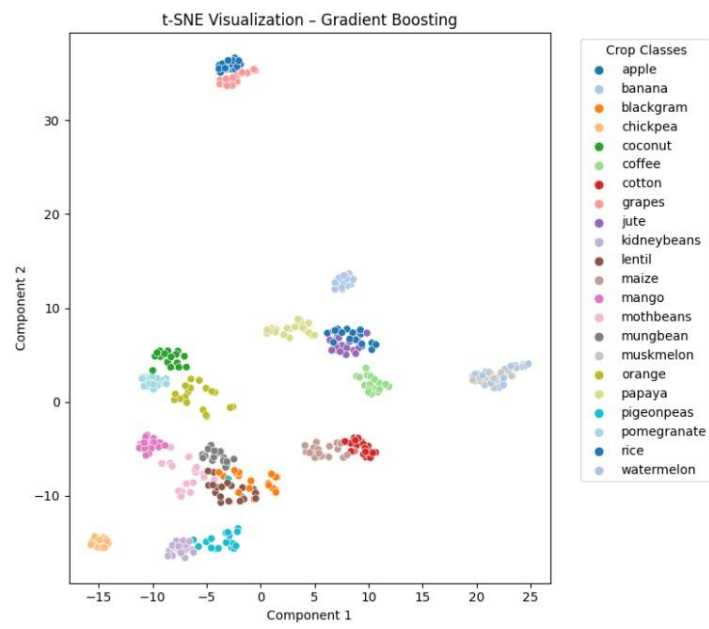


Figure 1.13: t-SNE of Gradient Boosting

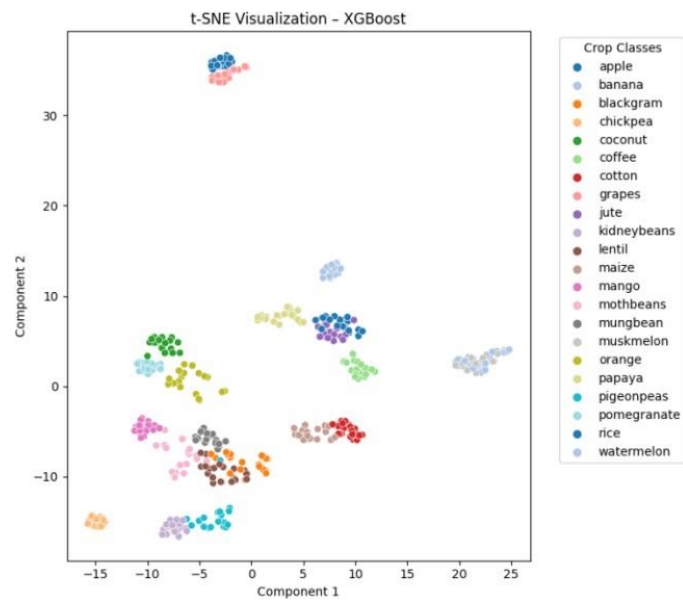


Figure 1.14: t-SNE of XGBoost

1.4.1 Cross Validation Results

Random Forest model results verify with 10-fold cross-validation achieved excellent performance, with an average accuracy of 0.9959 ± 0.0038 . The model also showed very high precision (0.9964), recall (0.9959), and F1-score (0.9959), indicating strong generalization and reliable crop prediction capability.

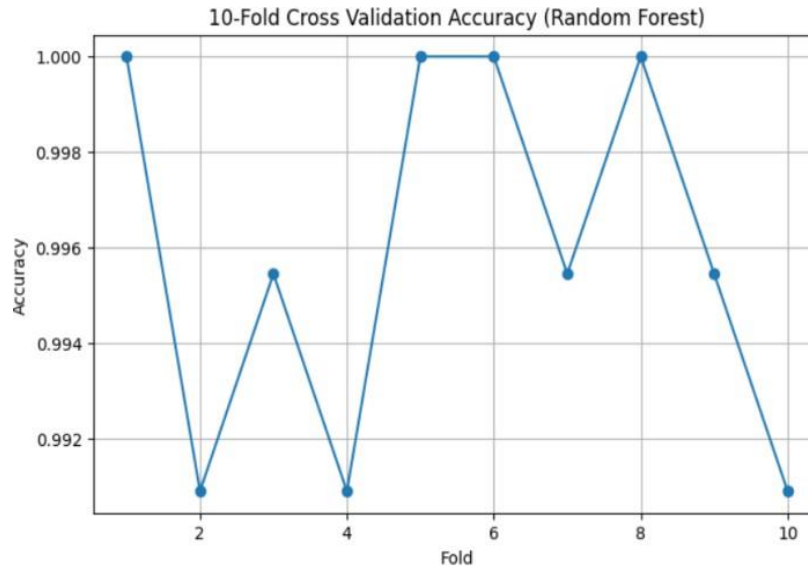


Figure 1.15: 10 Fold Cross Validation Report

C. Hugging Face Deployment and Real World Implementation

The most successful Random Forest model was deployed on the Hugging Face to build an interactive and real-time crop recommendation system. The web interface can be used to enter soil nutrient levels (N, P, K) and environmental conditions (temperature, humidity, rainfall, pH, etc.) and can be used to get real-time crop advice. The system deployed adheres to the identical preprocessing and feature engineering pipeline that is followed in model training, which guarantees that the predicted values remain consistent. The application does not give only one output but also comes up with a ranked list of the appropriate crops depending on the predicted probability score that allows informed decision-making in relation to practical constraints. The system can be made available to farmers and agricultural stakeholders without technical skills on how to use it since the deployment is done through the Hugging Face. It promotes seasonal planning, scenario-based analysis and effective utilization of resources.



Figure 1.16: Graphical User Interface of the Advanced Crop Recommendation System showing input parameters and the predicted output.

This picture is called Figure 16 and represents a user interface of an Advanced Crop Recommendation System. It is an evidence-based application that can assist farmers and researchers in determining the most appropriate crops that can be planted in certain environmental conditions.

Output: The analysis of these values has given the system the result or output recommendation of cultivation as Rice. In the center of the screen, there is a picture of rice and the text written in big fonts that say the name Rice.



Figure 1.17: Advanced Crop Recommendation System indicative of Top 3 Crop Predictions and Confidence Levels.

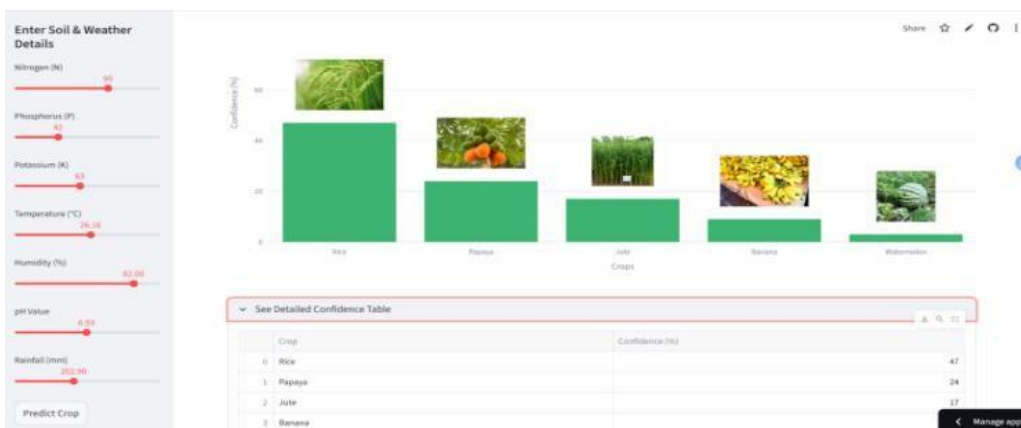


Figure 1.18: Crop Recommendation Dashboard

This Crop Recommendation Dashboard is a machine learning program that forecasts appropriate crops using soil nutrients and weather data including temperature and rainfalls. The bar chart is used to demonstrate the probability of success of the various crops where Rice has the highest confidence score of 47 thus making it the most appropriate crop.

Limitations and Future Work

The system is conditioned on a standard set of data and it does not take into account local, seasonal or real time environmental variation, or economic influence of crop decisions. The future work will be centered on incorporating real-time weather and soil sensor, extending the data to cover additional regions, and trying out more advanced models to enhance precision in agricultural work and make it more usable

1.5 CONCLUSION

The current paper has constructed a machine learning-based crop recommendation system using environmental and soil nutrient characteristics. Several classifiers were tested using sophisticated preprocessing and feature engineering methods. Random Forest showed the best test accuracy and F1-score and, thus, is a highly generalized and robust model. The better separation of classes of Random forest over other models was additionally confirmed by the use of confusion matrix and t-SNE visualizations. Depending on its performance, the model was implemented with the help of a streamlit based interface to give real-time crop recommendations. The suggested system is a promising and viable decision-support tool of precision agriculture, with the possibility of its future improvement based on larger and real-time datasets.

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